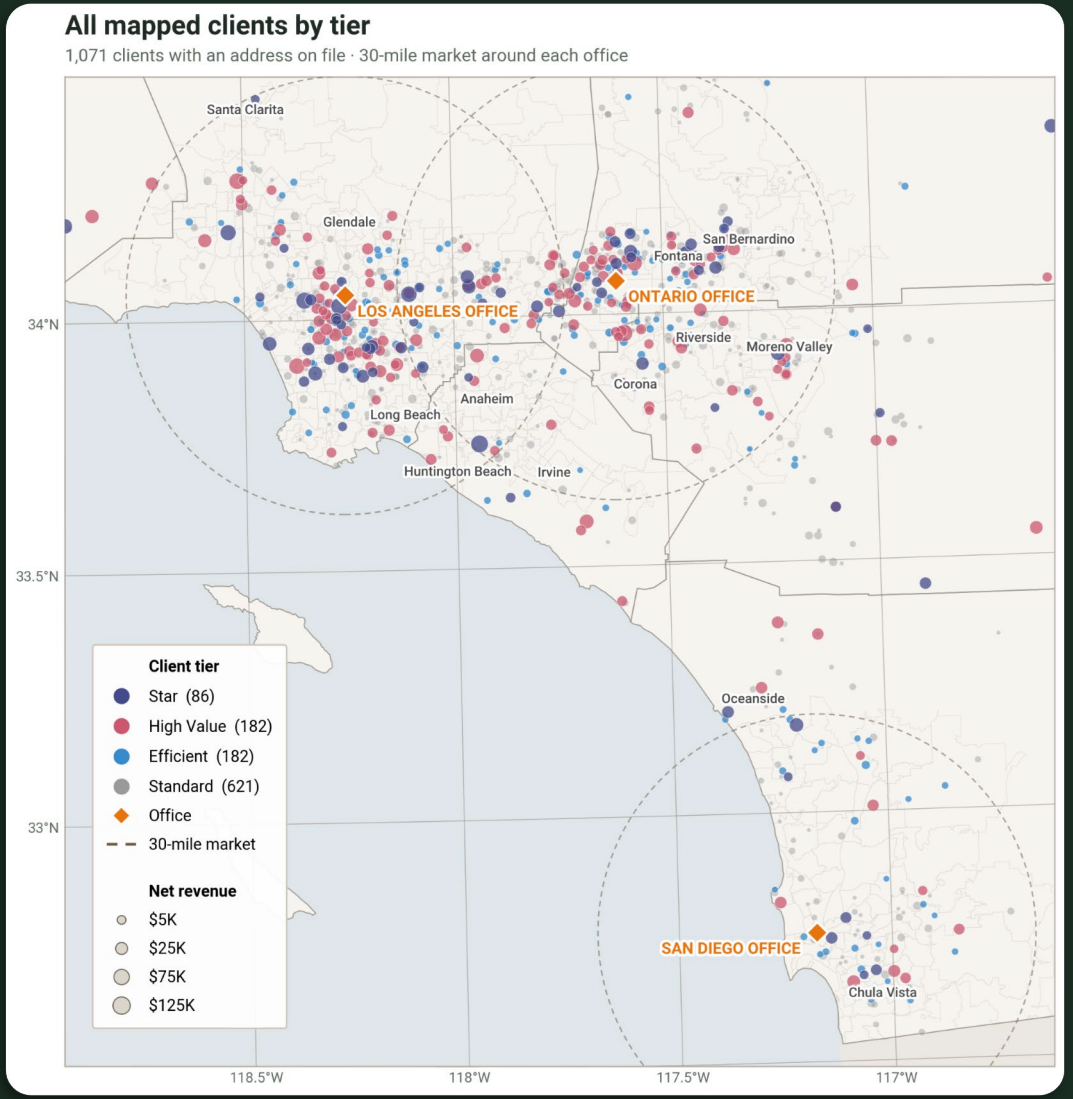
NACIS Practical Cartography Day

Where are your best clients?

Business analysis meets geospatial analysis: five years of law-firm records

Eugenie Huang & Reid Osborn
University of Wisconsin-Madison



The problem

A firm with detailed records and no geography

3
offices:
Los Angeles, Ontario, San Diego

2,066
cases,
2021–2026

1,874
clients

$9.57M
revenue

Three business questions

1. Who are the best clients?
2. Where do they live?
3. Where should the firm grow?

Our approach: both lenses at once

Business analysis finds **which** clients matter: revenue, hours, tiers.

Geospatial analysis finds **where** they are, and where more could be.

The four tiers

top revenue
High Value
182 clients
51% of revenue

top on both
Star
86 clients
33% of revenue

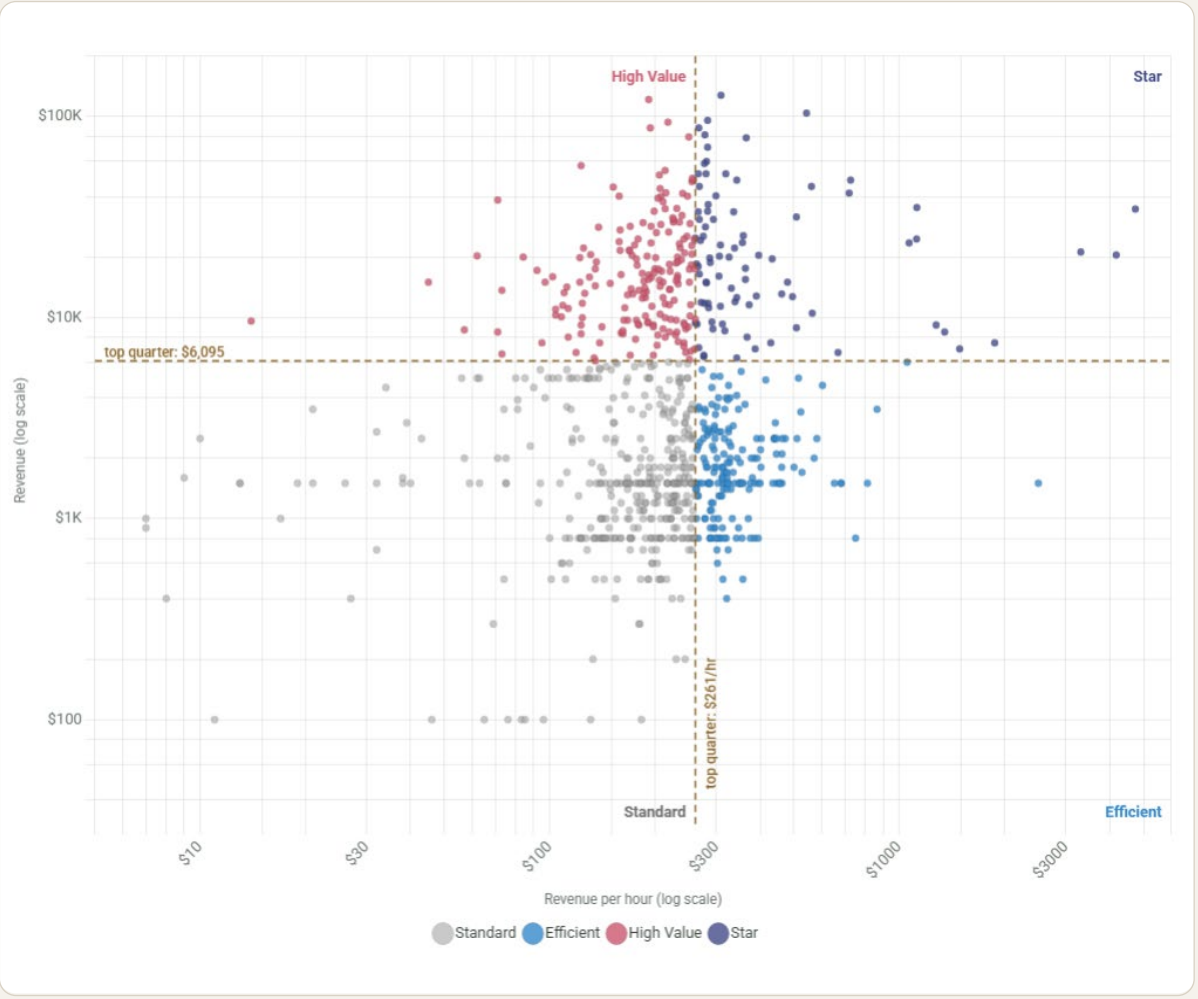
neither
Standard
621 clients
11% of revenue

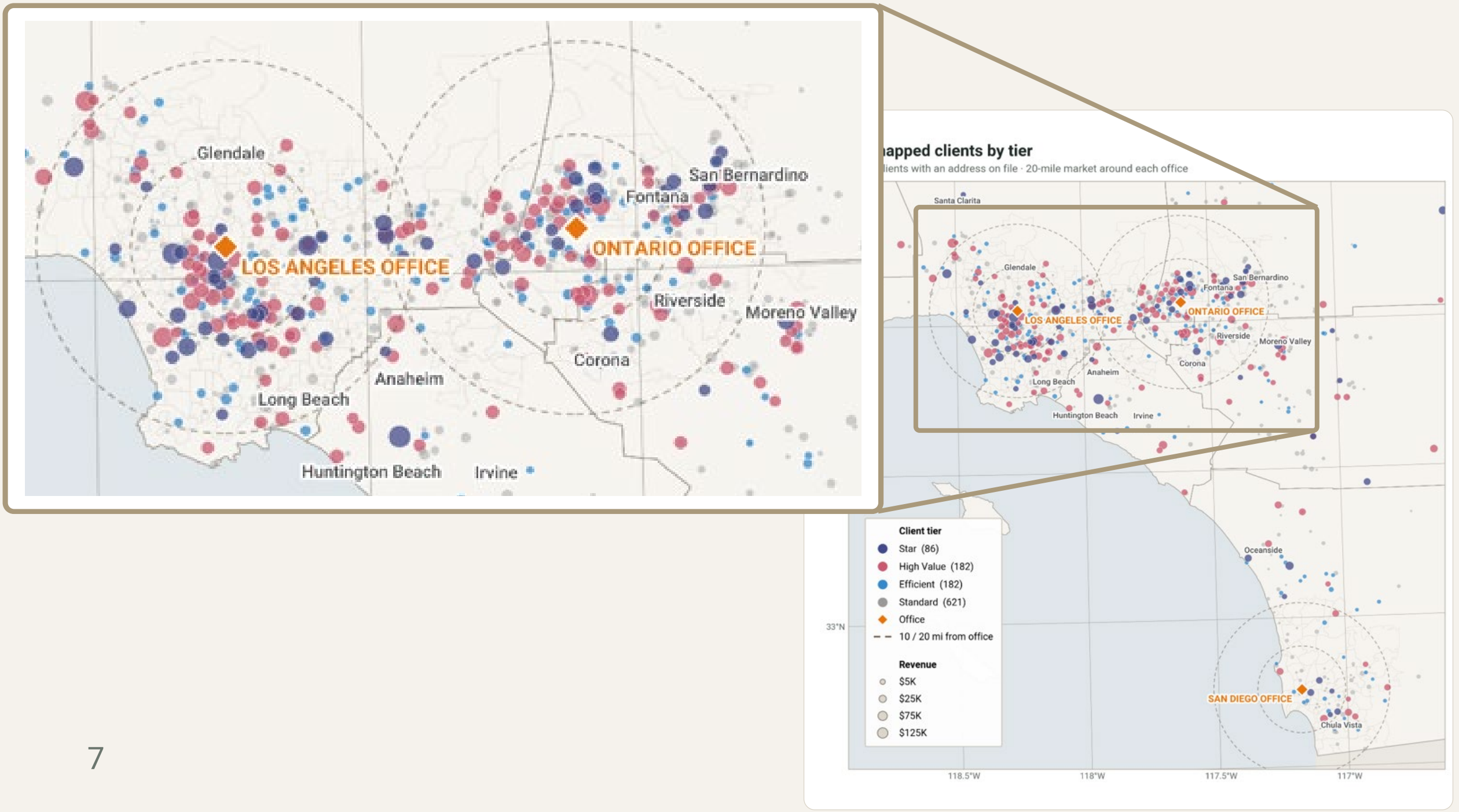
top hourly yield
Efficient
182 clients
5% of revenue

revenue

hourly yield

Cut-offs: **$6,095** and **$261/hr**.




apped clients by tier
lients with an address on file · 20-mile market around each office
Santa Clarita
Glendale
San Bernardino
Fontana
LOS ANGELES OFFICE
ONTARIO OFFICE
Riverside
Moreno Valley
Corona
Anaheim
Long Beach
Huntington Beach
Irvine
Oceanside
SAN DIEGO OFFICE
Chula Vista
Client tier
Star (86)
High Value (182)
Efficient (182)
Standard (621)
Office
10 / 20 mi from office
Revenue
$5K
$25K
$75K
$125K
33°N
118.5°W
118°W
117.5°W
117°W

Geog 574 final project, spring 2026 · Eugenie Huang & Reid Osborn

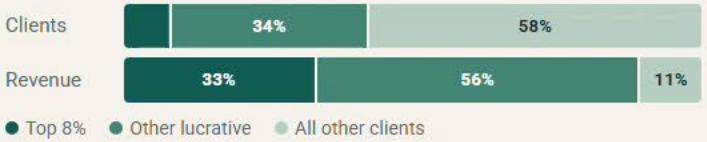
Where are your best clients?

We combined business analysis and geospatial analysis of five years of records from a Southern California civil law firm, to see who its most valuable clients are, where they live, and where the firm could look for more of them. Three findings stand out.

1

A small group of clients brings in most of the revenue

Share of mapped clients and of revenue



The top 8% and the other lucrative clients together are **42%** of mapped clients and bring in **89%** of revenue. The top 8% alone bring in a third.

2

Ontario reaches far more residents than the other two offices

Clients per 100,000 residents, within 30 miles



Yet the share of lucrative clients is similar or higher around Los Angeles (44%) and San Diego (46%) than around Ontario (40%).

3

The clearest room to grow is close to the offices

Clients per 100,000 residents, by miles from the nearest office



Reach beyond 20 miles is about half what it is within 10. Several populous ZIP areas within 20 miles of Los Angeles and San Diego have few clients today.

Read the full story

Key findings

Three statistics to remember

89%

of revenue comes from the **42%** of clients we consider lucrative

11.8

clients per 100,000 residents around Ontario, against **4.2** in LA and **3.3** in San Diego

2

Spanish-speaking clients in a San Diego market of **537,000** Spanish speakers

From the firm's CSVs to a StoryMap

Firm CSVs
contacts, matters, payments, hours

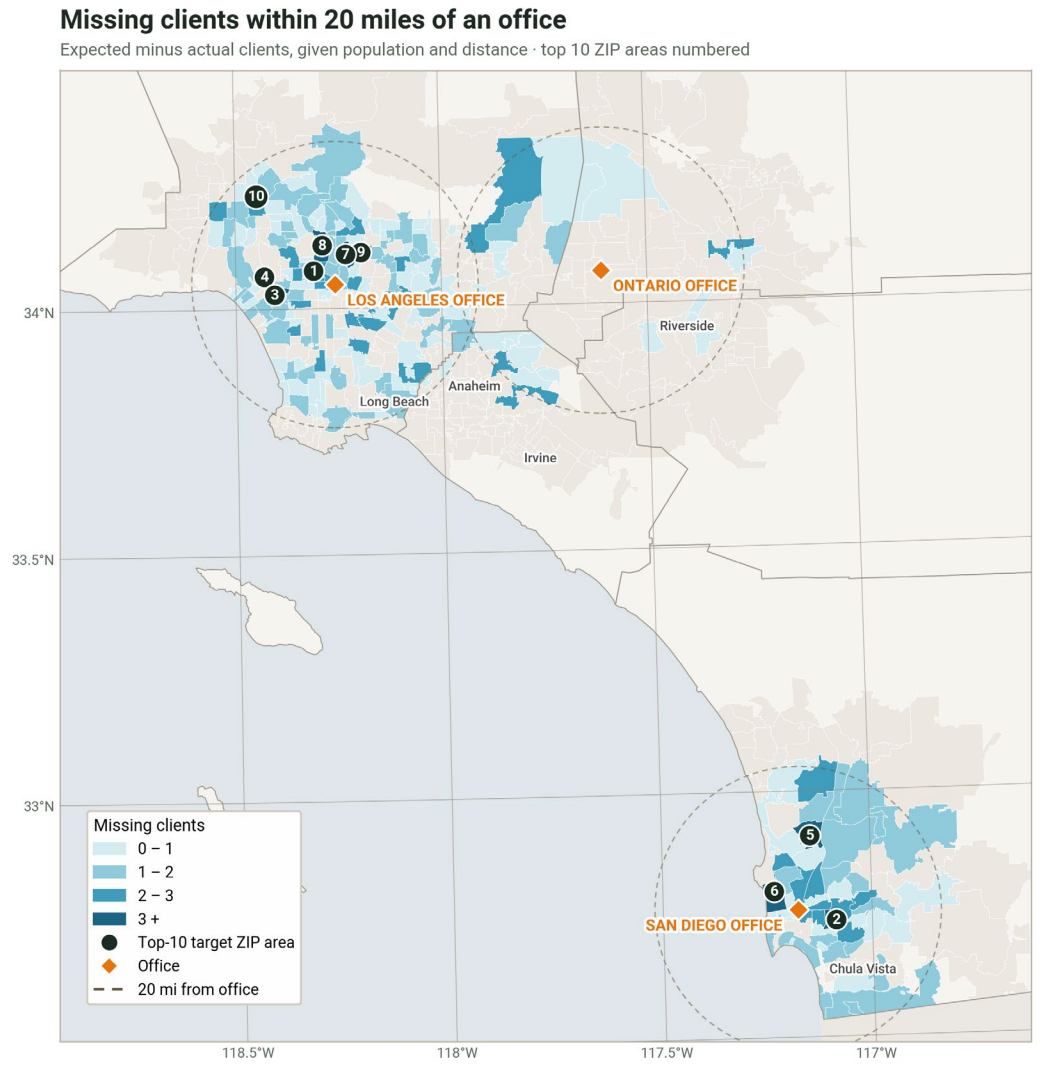
De-identify
personal details

Geocode
Google Geocoding API: address to point

PostGIS 3.6
4 firm tables, 2020 ZCTAs, ACS 2020-24

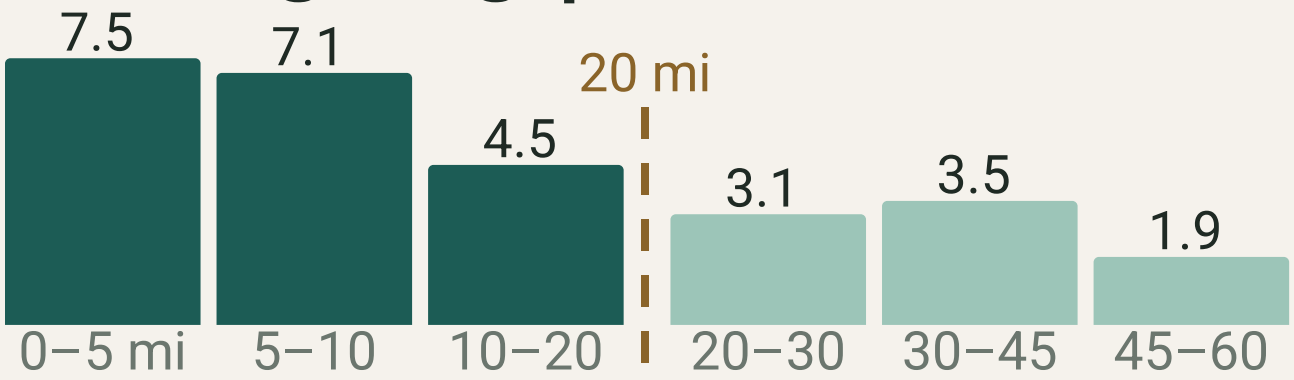
Python build
smoothing, Gi*, gap model, maps

StoryMap
HTML, JS, CSS, Chart.js



Methods · defining market areas

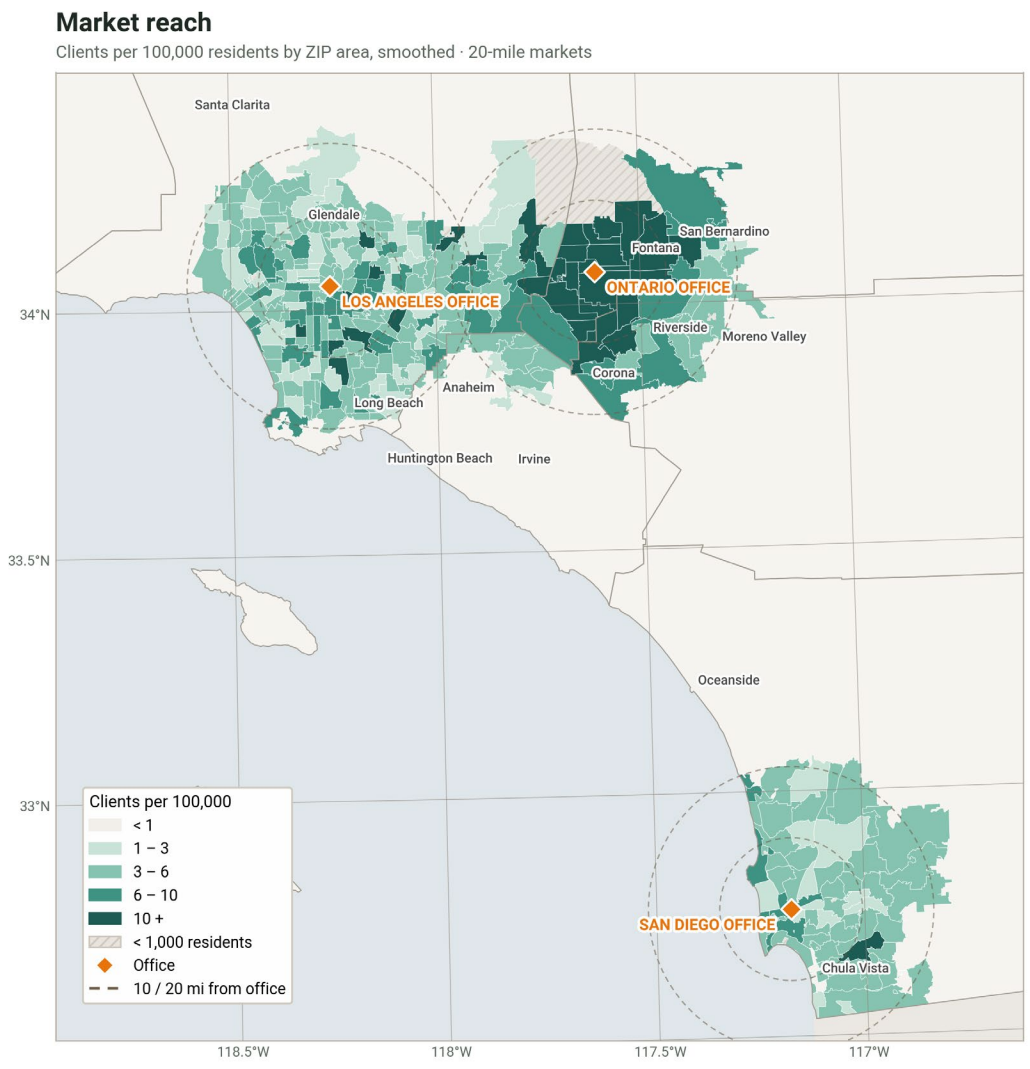
Modeling the gap



Clients per 100,000 residents by miles to the nearest office

expected = population × rate(distance band)
gap = expected − actual clients

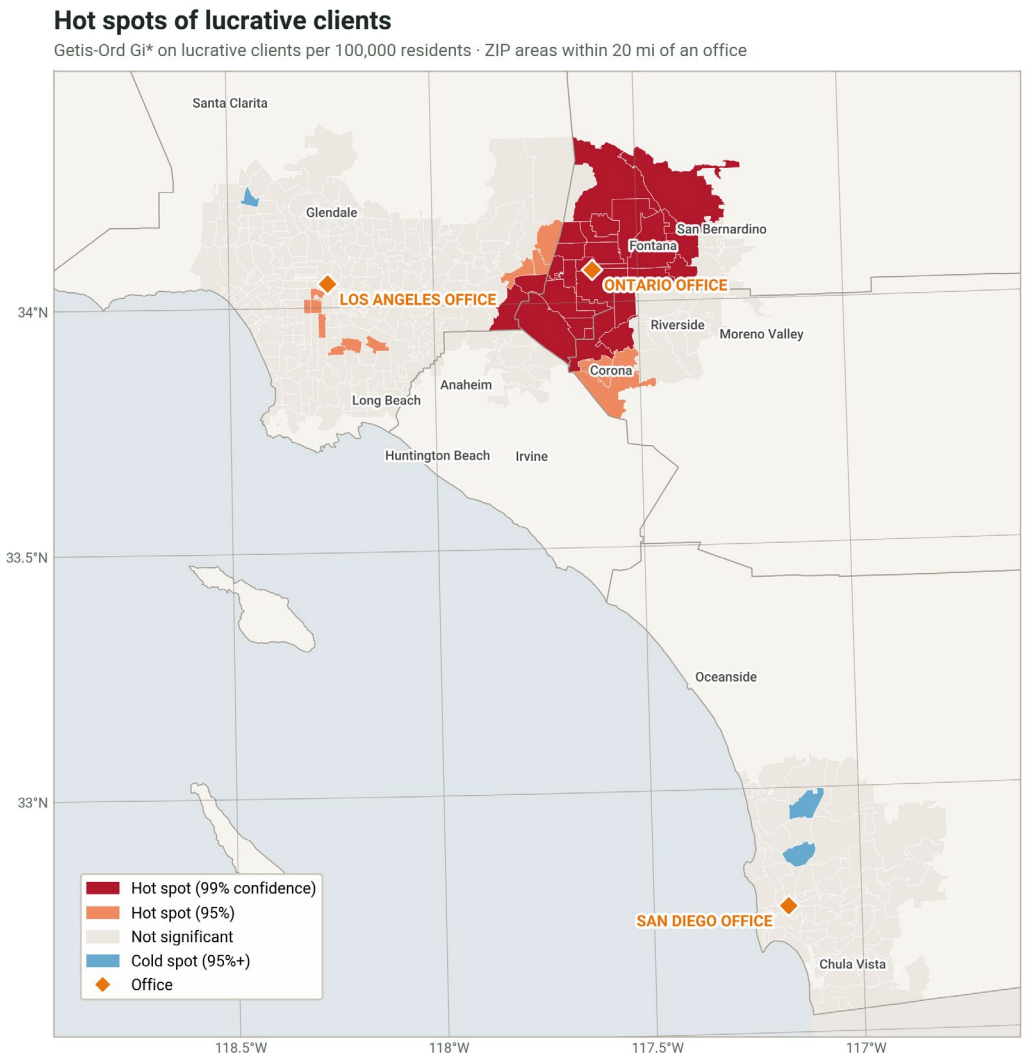
Within 20 miles: **≈196 missing clients around LA**, **≈69 around San Diego**, **≈18 around Ontario**. Same idea with Spanish speakers as the denominator gives the San Diego language gap.



Methods · from counts to rates

Empirical Bayes Smoothing

- Adjusts client rates based on population size.
- Reduces extreme rates in areas with small populations.
- Makes comparisons across ZCTAs more reliable.

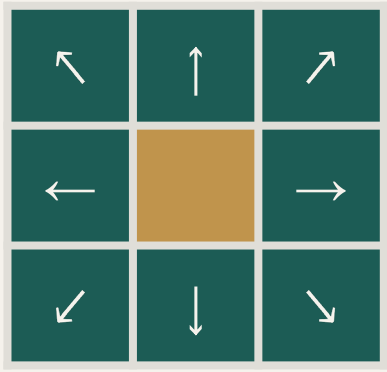


Methods · from counts to rates

Hot Spot Analysis (Getis-Ord Gi*)

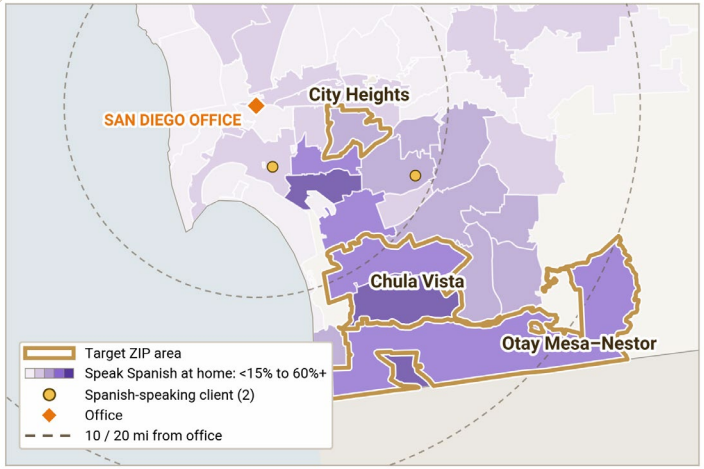
King/Queen Contiguity | 369 ZCTAs*

- **Ontario:** 32 hot spots
- **Los Angeles:** 7 hot spots
- **San Diego:** 0 hot spots



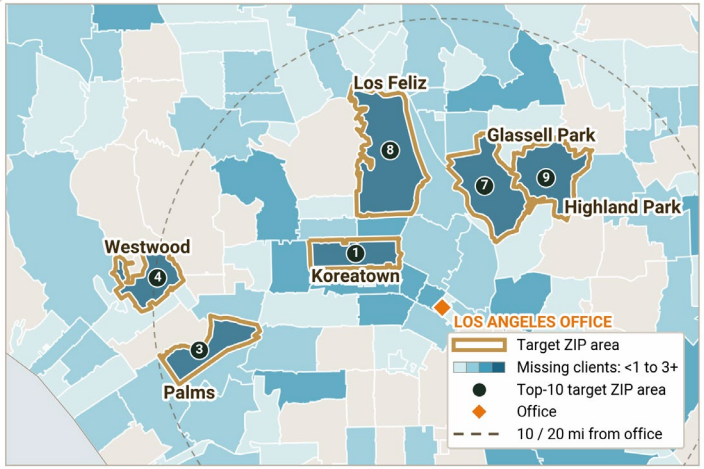
Recommendations

What we'd tell the firm



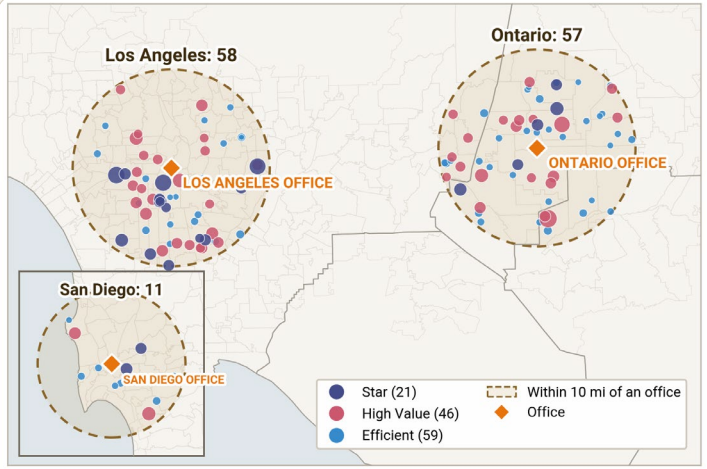
1 Spanish-language outreach in San Diego

Chula Vista, Otay Mesa–Nestor, City Heights



2 LA neighborhoods within 10 miles

Koreatown, Los Feliz, Glassell Park, Highland Park, Palms, Westwood



3 Past clients as referral sources

126 dormant lucrative clients within 10 miles of an office

Business analysis tells you who.
Geospatial analysis tells you where.
Together they tell the firm what to do next.



Thank you · Questions?

Eugenie Huang & Reid Osborn · UW-Madison
huang674@wisc.edu rkosborn@wisc.edu

https://cat-grep.github.io/SpatialBusinessIntelligence/